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Research papers

Nail penetration safety enhancement in semisolid-state batteries via composite solid-state electrolyte separators☆

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ABSTRACT

This study develops a composite solid-state electrolyte (SSE) separator for semisolid-state batteries, achieving a 450 Wh/kg lithium metal battery that passes stringent nail penetration tests. The SSE, composed of PEO-LLZTO coated on a polypropylene base, exhibits superior elongation and ionic conductivity enabling effective nail encapsulation during penetration. A custom steel nail with embedded thermocouples was used to monitor the maximum temperature at the penetration site of different pouch cells during the nail penetration test, providing more accurate measurements than surface-mounted thermocouples. A parallel circuit model quantification demonstrated significant differences in peak current and short-circuit internal resistance between liquid batteries and semisolid-state batteries. Semisolid-state batteries with 1.4 g/Ah electrolyte filling demonstrated safety across capacities (3.2–7.5 Ah), whereas liquid batteries failed. The posttest microscope images confirmed the role of SSE in reducing the peak splitting height and fracture area of the electrodes. This work provides a methodological framework for nail penetration analysis, highlighting SSE separators as a solution to balance energy density and safety.

1. Introduction

Lithium-ion batteries (LIBs), which are celebrated for their exceptional energy density (200–300 Wh/kg), extended cycle life (>2000 cycles), and environmental sustainability [1–3], have revolutionized applications spanning electric vehicles, grid-scale energy storage, and portable electronics. However, their widespread adoption is tempered by inherent safety risks: conventional LIBs employing flammable carbonate electrolytes (flash point of ~32 °C) exhibit thermal runaway vulnerabilities under mechanical abuse [4,5]. Solid-state batteries (SSBs) have emerged as paradigm-shifting solutions by replacing liquid electrolytes with inherently nonflammable solid electrolytes (e.g., oxides and sulfides) [6]. This innovation not only eliminates combustion risks but also unlocks the potential of lithium metal anodes, theoretically doubling the energy density [7]. Consequently, SSBs are positioned as the gold standard for next-generation energy storage, simultaneously addressing safety and performance bottlenecks that have historically constrained LIB development.

While solid-state batteries (SSBs) represent a significant advancement in safety over conventional lithium-ion batteries, recent experimental studies and comprehensive safety assessments demonstrate that they remain susceptible to failure mechanisms, particularly abnormal interfacial impedance and lithium dendrite penetration, which can compromise electrochemical performance and induce catastrophic short circuits, ultimately leading to thermal runaway [8,9]. To address these limitations, semisolid-state batteries have emerged as pragmatic hybrid solutions that combine high-ionic-conductivity solid-state electrolyte separators with minimal liquid electrolytes or wetting agents to optimize electrode interface compatibility [10–12]. However, this transition introduces a critical trade-off: immature fabrication processes or sub-optimal chemical systems may inadvertently reduce the energy density when traditional separators are replaced with solid-state alternatives

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[13–15]. To reconcile safety and energy density, researchers have employed multiple strategies: high-capacity electrode materials the adoption of Ni93 cathodes and Si/C anodes or lithium metal to increase the specific capacity [16–18]; process innovations the use of dry-process thick electrodes to increase active material loading; in situ polymerization to minimize the electrolyte volume [19]; and separator engineering the development of ultrathin, uniform composite solid-state electrolyte separators to reduce weight while maintaining ionic conductivity [20–23]. The design of these composite separators must rigorously balance electrochemical performance (e.g., ionic conductivity, interfacial stability), manufacturing feasibility, and energy density optimization to ensure that both safety and performance are achieved.

Among the most rigorous battery safety evaluation protocols, nail penetration testing represents a particularly stringent abuse scenario that has become a critical focus in battery failure research [24–28]. The rapid onset of thermal runaway induced by this mechanism presents significant challenges for experimental observation, as the process unfolds too quickly for comprehensive data capture. In response to these challenges, researchers have extensively investigated the mechanisms of nail-induced internal short circuits and subsequent thermal runaway. These studies have included material improvements, including enhanced thermal stability of cathode/anode materials and optimization of separator/electrolyte flame retardancy [29,30], as well as cell design innovations such as current collector optimization for improved current conduction and heat dissipation, active material formulation adjustments to enhance interfacial resistance, and ceramic surface coatings for elevated thermal resistance [31,32]. Other research efforts also include the development of methods for monitoring internal temperature, such as integrating sensors into the battery to monitor the thermal runaway process [33,34], which is crucial for decomposing and understanding the stages of thermal runaway. Additionally, comprehensive testing protocols have been developed to investigate variables, including test conditions and state-of-charge (SOC) effects, accompanied by detailed cell safety failure analysis [35,36]. Advanced thermo-electrical coupled simulation models have also been employed to study thermal runaway processes at both the cell and pack levels [37,38]. Mechanistic studies reveal that the dominant heat source in nail-induced thermal runaway, contributing approximately 80% of total heat generation, originates from the interaction between the positive current collector (Al) and the fully charged negative electrode [39]. This finding underscores the separator's critical function in preventing short circuits between the Al foil and the negative electrode, highlighting its pivotal role in battery safety [40].

Solid-state electrolyte separators present a promising solution for addressing critical battery safety challenges. The development of solid-state batteries hinges fundamentally on the creation of solid-state electrolytes that combine high stability with exceptional ionic conductivity to replace conventional liquid electrolytes [41]. Composite solid-state electrolytes, which integrate polymer matrices with oxide-based ion-conducting channels, synergistically leverage the complementary advantages of both material systems. For example, oxide-based solid electrolytes such as LLZTO have a unique three-dimensional ion conduction channel, whereas LATP variants exhibit both thermal resilience and relatively high ionic conductivity. Polymer-based systems, including PEO and PVDF separators, offer distinct advantages because of their superior elongation rates and tensile strength, respectively, and Vi-PDMS polymer separators stand out for their remarkable toughness and elasticity. These polymer membranes, characterized by their inherent flexibility, high elongation capabilities, and ionic conductivity, may provide key insights for enhancing overall battery safety [42,43]. A central objective of this investigation is to evaluate whether such composite solid-state electrolyte separators can effectively mitigate the severity of internal short circuits, thereby improving both the safety performance metrics and the electrochemical characteristics of semisolid-state battery systems.

Owing to the rapid and transient nature of battery nail penetration events, current methodologies for process decomposition remain insufficiently developed [44–48]. To address this critical knowledge gap, this study establishes a comprehensive research framework for analyzing internal short-circuit heat generation and process dynamics during nail penetration. A key innovation involves the use of specially designed stainless steel nails equipped with embedded temperature sensors, enabling precise measurement of internal temperature variations rather than relying solely on surface or tab temperature readings. The experimental system integrates a dual-battery parallel circuit model with a Hall-effect coil-based current measurement architecture, creating a novel data acquisition platform for detailed process characterization. The primary objectives of this study are twofold: (1) to develop advanced solid-state electrolyte separators specifically tailored for soft-pack battery (pouch cell) applications and (2) to establish a groundbreaking research methodology for nail penetration process analysis. This approach systematically quantifies critical operational parameters, including temperature evolution, voltage drop characteristics, current fluctuations, internal resistance changes, and power variations throughout the penetration event. By correlating these key process parameters with heat generation and discharge power characteristics, the methodology provides mechanistic insights into nail penetration phenomena while offering a versatile analytical framework applicable to diverse material systems and battery configurations. Ultimately, this research decomposition strategy could serve as a foundational methodology for safety optimization across various battery architectures.

2. Experimental

2.1. Fabrication of PEO/LLZTO composite solid-state electrolyte separators

A composite solid-state electrolyte was synthesized by integrating polyethylene oxide (PEO: Mv ≈ 600,000; Adamas-beta) with lithium lanthanum zirconium tantalum oxide (LLZTO, Li6.4La3Zr1.4Ta0.6O12, D50 = 3 μm; GEN LU ADVANCE) through a three-step process. The preparation began by dissolving PEO powders in acetonitrile to form a homogeneous solution (10% solid content), followed by the controlled dispersion of LLZTO particles into the PEO matrix at mass ratios ranging from 30% to 70%, and the ratios of PEO to LLZTO were 3:7 (P1), 4:6 (P2), 5:5 (P3), 6:4 (P4), and 7:3 (P5). The resulting composite slurry was then uniformly coated onto a 7 μm polypropylene (PP) substrate, yielding a precisely controlled solid-state electrolyte layer with a thickness of 7 μm. This PEO/LLZTO composite separator was employed in semisolid-state batteries, whereas conventional liquid lithium-ion batteries utilized a 14 μm commercial PP base separator (Shenzhen Senior Technology Material Co., Ltd.) without any coating. The method demonstrates a systematic approach to balance ionic conductivity and mechanical stability for advanced battery applications.

2.2. Fabrication and characterization of high-energy-density pouch cells with PEO/LLZTO solid-state electrolytes

The pouch cell assembly was conducted in a controlled environment with a − 50 °C dew point to ensure moisture-sensitive material stability. The cathode slurry was prepared by dispersing LiNi0.93-Co0.05Mn0.02O2 (Ni93, Ronbay New Energy Technology Co., Ltd.), carbon nanotubes (CNTs, Jiangsu Cnano Technology Co., Ltd.), and polyvinylidene fluoride (PVDF, Solvay S.A.) in *N*-methyl-2-pyrrolidone (NMP) at 97.5:1.25:1.25 wt%, whereas the anode slurry consisted of SiO_x/C 550 (Tianmulake Excellent Anode Materials Co., Ltd.), Super P (SP, C65, TIMCAL), and polyacrylic acid-styrene butadiene rubber (PAA from Sichuan Indigo Technology Co., Ltd. +SBR from ZEON) binder in deionized water with 96:2:2 wt%. Both slurries were sequentially coated onto current collectors, roll-pressed, and slit into precise dimensions before being stacked with the PEO/LLZTO solid-state electrolyte separator (prepared in section 2.1). The electrode assembly underwent vacuum encapsulation, electrolyte injection (EC: DEC = 1:1 with 1 M $LiPF_6$, Guotai Huarong New Materials Co., Ltd.), formation cycling, and capacity grading to finalize cell production. The pouch cells formed at 45 °C under a pressure of 300 kgf. The formation protocol involved charging at 0.05C to 3.75 V first, followed by charging at 0.1C to 4.25 V, and finally 0.1C discharging to 2.5 V.

Four pouch cell configurations were designed with dimensions of 80 × 120 mm: (1) single-layer cells achieving 0.35 Ah capacity, (2) 3-layer cathode/4-layer anode cells delivering 3.2 Ah (340 Wh/kg), (3) 8-layer cathode/9-layer anode cells reaching 7.5 Ah (380 Wh/kg), and (4) lithium metal anode (60 μm) variants matching the same structure of (3) but achieving 450 Wh/kg energy density at 7.5 Ah. This modular design enables systematic evaluation of the impact of electrode configuration on electrochemical performance.

2.3. Comprehensive electrochemical characterization and safety evaluation of PEO/LLZTO-based battery systems

Electrochemical characterization was systematically performed using both coin cell and pouch cell configurations. The coin cells were assembled in a glove box (MBRAUN UNIlab, Shanghai) under an inert atmosphere. The prepared SSE separator and PP separator were punched into discs with a diameter of 16 mm. For the ionic conductivity of the coin cells, stainless steel spacers (diameter 16 mm, thickness 0.5 mm) were used for clamping and assembly. To determine the ion transfer number of the coin cells, lithium sheets (diameter 15.6 mm, thickness 0.45 mm) were used for clamping and assembly. Both types of coin cells were infiltrated with 100 μL of 1 M $LiPF_6$ electrolyte EC/DEC (1:1 wt) (Guotai-Huarong New Chemical Materials Co., Ltd. Zhangjiagang) and then sealed (under 450 kgf pressure for 10 s). The cells were left to stand at room temperature for 5 h before testing. The ionic conductivity of the solid-state electrolyte (SSE) separator was quantified through electrochemical impedance spectroscopy (EIS) on SS||SSE separator||SS and SS||PP separator||SS coin cells (SS: stainless steel spacers) via an electrochemical workstation (Zahner XC, Germany) with 5 mV AC perturbation across the 10ˆ–2–10ˆ5 Hz frequency range, which was calculated on the basis of eq. (1):

$$\sigma = \frac{L}{R \times S} \tag{1}$$

where σ is used to indicate the Li-ion conductivity of either the base separator or SSE separator, whereas L, R and S represent the thickness of the separator, the bulk impedance as measured by EIS and the effective contact area between the two blocked electrodes, respectively.

The number of Li ions transferred was determined via the Bruce–Vincent method on Li||SSE separator||Li and Li||PP separator||Li symmetrical cells under 10-mV polarization until a steady-state current was achieved. The calculation formula was derived from eq. (2):

$$t_{Li^+} = \frac{I_s \times (\Delta V - I_0 \times R_0)}{I_0 \times (\Delta V - I_s \times R_s)} \tag{2}$$

where I_0 and I_s correspond to the initial and stabilized currents observed during the polarization process. Concurrently, R_0 and R_s represent the interfacial resistances at the onset and the conclusion of the polarization process, respectively, reflecting the conditions before and after the process.

Pouch cell evaluation involved cycling at a 1/3C rate (2.5–4.25 V) using battery testers (NEWARE, Shenzhen) at 25 °C, complemented by EIS characterization (Zahner XC, 10,000 Hz–100 MHz, 200 mA perturbation). The safety assessment included nail penetration tests conducted under standardized conditions: fully charged cells (4.25 V, 0.05C cutoff) were subjected to Φ3 mm steel nail penetration at 80 mm/s, with the temperature maintained at 25 °C throughout the test. This multiscale evaluation protocol provides comprehensive insights into both the electrochemical performance and safety characteristics of the PEO/LLZTO-based battery system.

2.4. Advanced characterization of solid-state electrolyte separators for lithium-ion batteries

The microstructural and mechanical properties of solid-state electrolyte (SSE) separators were systematically characterized via complementary analytical techniques. Field-emission scanning electron microscopy (FE-SEM, JEOL, Japan) was employed with the following optimized parameters: 10 kV accelerating voltage, 6–10 mm working distance, and magnification ranging from 500× to 10,000×. Sample preparation was conducted in a glove box using conductive adhesive-coated holders to ensure contamination-free transfer to the vacuum environment.

Comprehensive physical property analysis included thickness and area density measurements via a micrometer and five-decimal analytical balance (Metrohm, Switzerland), mechanical testing via tensile equipment (China Machinery Test Equipment Co., Ltd.), and gas permeability evaluation via a dedicated tester (Guangzhou Biaoji Packaging Equipment Co., Ltd.). Superresolution microscopy (Keyence, Japan) was further utilized to analyze the electrode microstructure, particularly the electrode split peak structure and penetration-induced morphological changes. This multitechnique approach provides thorough insights into the separator performance characteristics critical for battery optimization.

3. Results and discussion

3.1. Comparative physicochemical analysis of solid-state electrolyte separators for lithium-ion batteries

The electrochemical performance of SSE separators was systematically compared with that of conventional base separators through comprehensive characterization. As shown in Fig. 1(a), the ionic transference number of the SSE separator reached 0.94, which was markedly greater than the base separator's value of 0.86. Conductivity measurements revealed that the SSE separator exhibited superior ionic conduction efficiency, with the highest conductivity of 3.32×10^{-4} S/cm [49], compared with 2.59×10^{-4} S/cm for the base separator (Fig. 1(d)).

Scanning electron microscopy (SEM) analysis demonstrated distinct microstructural differences: the base separator showed uniform porosity indicative of mature processing, whereas the SSE separator displayed a uniform poly(ethylene oxide) (PEO) polymer wrapping around LLZTO crystal surfaces (Fig. 1(b-c)). The LLZTO particles exhibited controlled particle sizes of 3–5 μm, with magnified images revealing surface transformation from glossy to matte white, confirming homogeneous polymer adhesion (Fig. 1(e)-(f)). Thickness measurements confirmed that the total thickness of the SSE separator was 14 μm, corresponding to a 7 μm coating layer, demonstrating precise control over interfacial engineering. These results collectively highlight the enhanced electrochemical performance of the solid-state electrolyte system.

3.2. Optimization of the PEO/LLZTO ratio in solid-state electrolytes for enhanced battery performance

The electrochemical performance of solid-state electrolytes was systematically evaluated through formulation screening of PEO/LLZTO composites. A series of pouch cells were fabricated with the PEO content progressively increasing from 30% to 70% in 10% increments (P1–P5). The ionic conductivities of the five schemes were 1.26×10^{-4}, 2.09×10^{-4}, 3.32×10^{-4}, 3.01×10^{-4}, and 2.81×10^{-4}, respectively, and P3 had the highest ionic conductivity. Accordingly, cycling performance analysis revealed that the P3 formulation with 50% PEO content exhibited exceptional stability, maintaining 88.4% capacity retention after 500 cycles at a 1/3C rate (Fig. 2(a)-(c)).

Electrochemical impedance spectroscopy (EIS) measurements

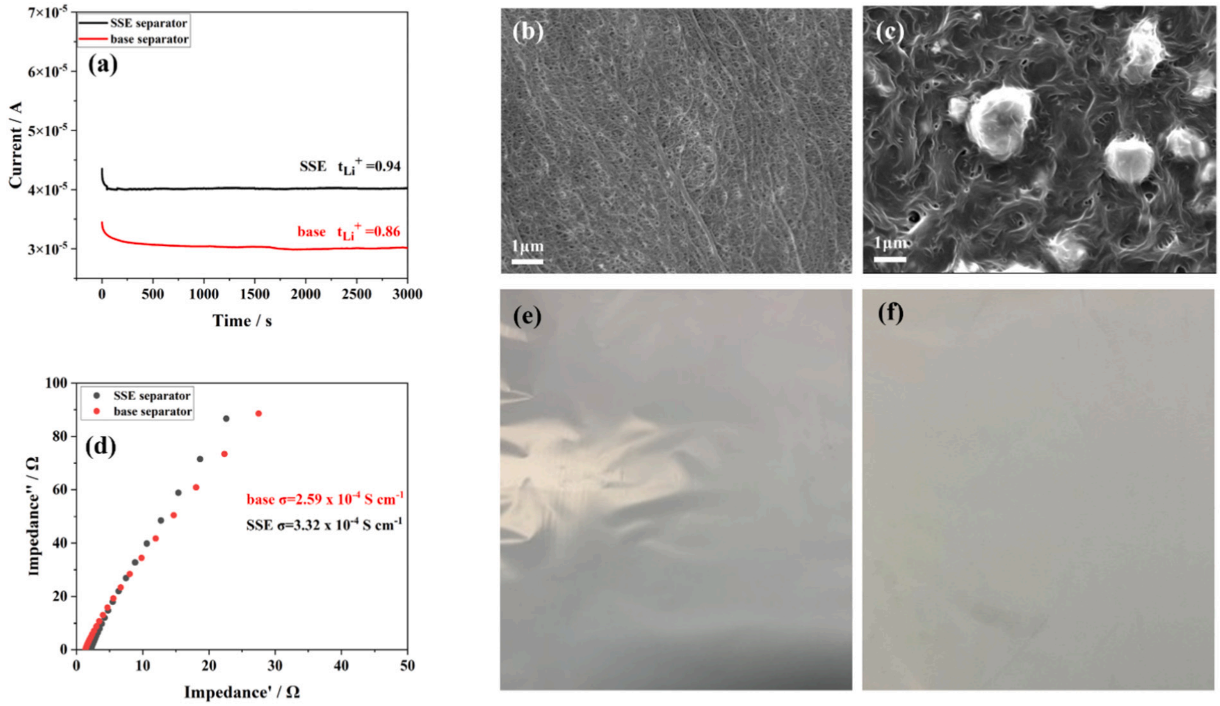

Fig. 1. Comparative characterization of solid-state electrolyte (SSE) and base separators: (a) Lithium-ion transference numbers; (b, c) Scanning electron microscopy images of base separator (b) and SSE separator (c); (d) Ionic conductivity profiles; (e, f) Macroscopic views of magnified base separator (e) and SSE separator (f).

provided mechanistic insights into the performance variation. Initial characterization revealed consistent ohmic resistance (Rs) across all formulations prior to cycling (Fig. 2(d)), confirming uniform cell assembly. Postcycling analysis revealed minimal Rs changes but significant differences in the charge transfer resistance (Rct), with the P3 formulation demonstrating the lowest Rct value (Fig. 2(e)). This observation correlates directly with its superior cycling stability, confirming the optimal balance between polymer flexibility and ceramic conductivity achieved at the 50%–50% PEO/LLZTO ratio. These findings provide critical guidance for solid-state electrolyte formulation optimization.

3.3. Electrochemical performance validation of semisolid-state pouch cells for lithium-ion batteries

Building upon the optimized formulation identified in previous studies (Scheme P3), we developed multilayer pouch cells with a designed capacity of 3.2 Ah to systematically compare liquid- and semisolid-state battery configurations under varying electrolyte conditions. As shown in Fig. 3(a), the liquid battery with a conventional separator demonstrated the highest initial discharge capacity (3.55 Ah), whereas the semisolid-state counterpart achieved 3.42 Ah at the equivalent electrolyte filling coefficient (defined as the ratio of the amount of liquid electrolyte injected to the battery capacity, g/Ah). Notably, when the filling coefficient was reduced from 2.0 to 1.5 g/Ah, the semisolid-state battery maintained excellent capacity consistency, with only a marginal decrease to 3.38 Ah. The long-term cycling performance revealed more pronounced advantages for the semisolid-state system. After 200 cycles at a 1/3C rate (Fig. 3(b)), the semisolid-state cell with 1.5 g/Ah exhibited superior capacity retention (89.8%) compared with the liquid battery with 2.0 g/Ah (86.7%). These results demonstrate that despite its lower initial capacity, the semisolid-state configuration offers enhanced cycling stability, particularly under reduced electrolyte conditions. These findings validate the practical viability of our optimized solid-state electrolyte formulation for high-performance battery applications.

Building upon previous findings, we developed advanced semisolid-state batteries with 8-layer positive electrode/9-layer negative electrode configurations to investigate electrolyte optimization strategies. A systematic electrolyte gradient experiment was conducted with injection coefficients of 1.8, 1.6, and 1.4 g/Ah. Remarkably, the semisolid-state battery demonstrated exceptional capacity stability across all the tested coefficients, maintaining consistent performance at approximately 6.9 Ah (Fig. 3(c)), confirming its robustness when the injection coefficient exceeded 1.4. Temperature-dependent studies revealed distinctive thermal responses: at 45 °C, the semisolid-state battery exhibited a capacity increase from less than 7 Ah to nearly 7.5 Ah, demonstrating higher temperature sensitivity than the stable output of the liquid battery of 7.34 Ah. Further energy density optimization was achieved by replacing the negative electrode with 60 μm lithium metal, resulting in a peak capacity of 7.64 Ah at 45 °C (Fig. 3(d)) and a calculated energy density of 450 Wh/kg. Comprehensive nail penetration tests were performed across different capacities and electrolyte conditions, systematically validating the superior safety characteristics of the solid-state electrolyte separator. These results collectively establish the semisolid-state configuration as a promising high-performance battery platform with balanced electrochemical properties and enhanced thermal tolerance.

3.4. Advanced thermal monitoring and safety evaluation of semisolid-state batteries through nail penetration testing

To systematically evaluate the thermal behavior during mechanical abuse, a custom-designed stainless steel monitoring nail was developed with embedded thermocouples for precise temperature measurement. As illustrated in Fig. 4(a)-(d), the 3 mm-diameter nail with a 45° tip angle incorporated six thermocouples (T1–T6) at strategic positions: T1–T4 at the cell tabs and surfaces, T5 on the inner nail wall, and T6 5 mm from the tip (Fig. 4(c)-(d)). The physical characterization of the instrument is shown in Fig. 4e.

Comparative analysis of the temperature profiles and voltage responses (Fig. 5(a)-(b)) revealed distinct thermal characteristics. During

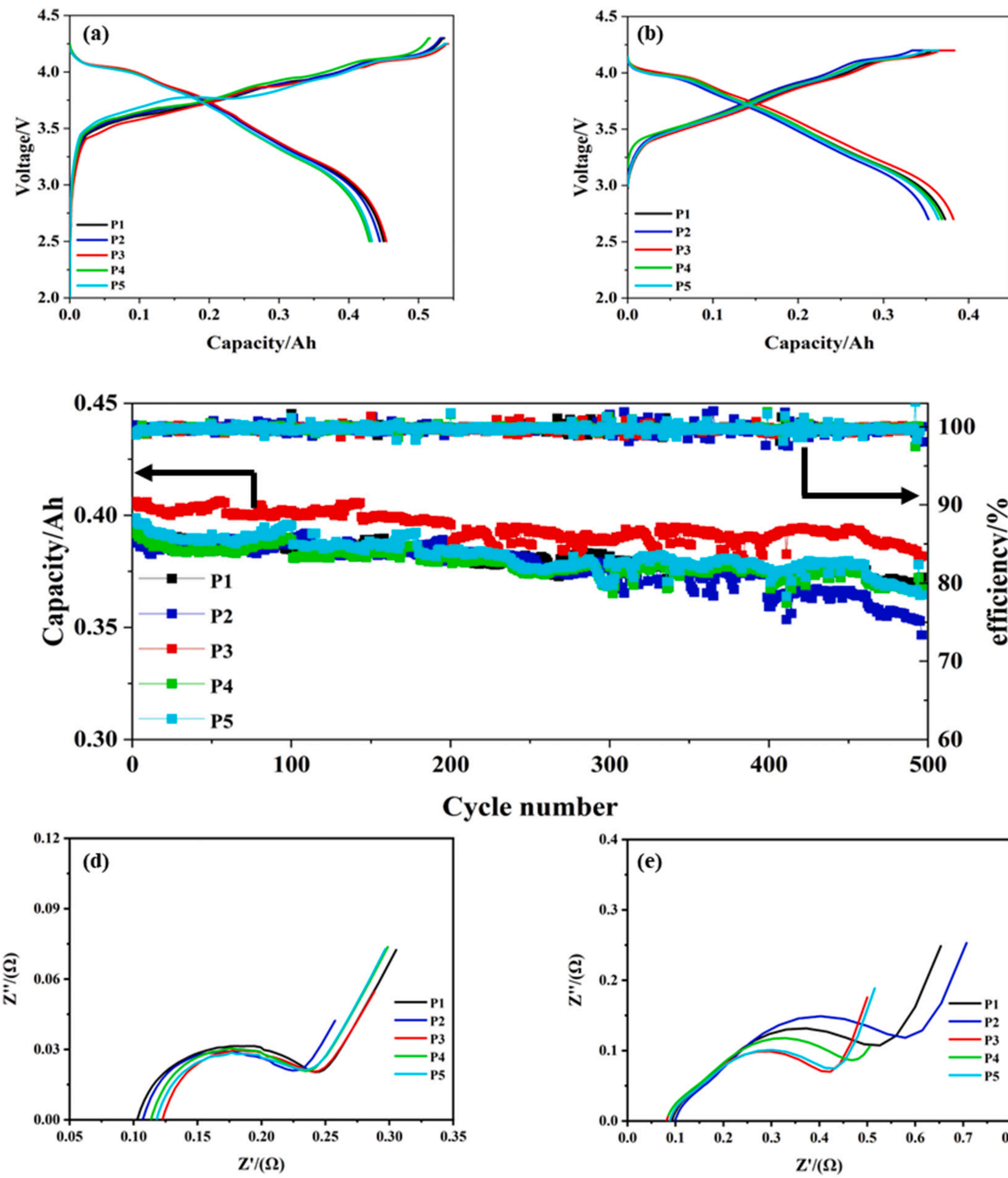

Fig. 2. Electrochemical performance comparison of different semi-solid-state pouch cell schemes: (a) Initial state; (b) After 500 cycles; (c) 1/3C cycling curves; (d) Initial EIS spectra; (e) EIS spectra after 500 cycles.

nonthermal runaway conditions, the T6 tip sensor recorded a peak temperature of 75 °C, which was significantly higher than those of T5 (60 °C) and T1 (31 °C). The number of thermal runaway events dramatically increased, with T6 approaching 500 °C and T5 reaching 392 °C. The voltage dynamics showed fundamental differences: the liquid batteries collapsed from 4.28 V to 0 V within 4.9 s, whereas the semisolid-state cells demonstrated remarkable stability, with the voltage decreasing from 4.19 V to 4.01 V in 1.5 s before recovering to 4.06 V within 0.6 s.

The data results from Table 2 and Supplementary Material S1 demonstrated that semisolid-state cells employing SSE separators exhibited significantly improved nail penetration safety compared with liquid electrolyte batteries with base separators—all liquid cells experienced thermal runaway regardless of capacity—whereas semisolid-state cells improved nail penetration safety at reduced injection coefficients. Notably, the lithium-metal configuration achieved an energy density of 450 Wh/kg, which passed safety tests at 1.4 g/Ah boundary conditions. In 3/4-layer pouch cells, regardless of the liquid injection coefficient, a clear difference in nail penetration outcomes was observed between cells with and without solid-state electrolyte separators. In 8/9-layer pouch cells, especially those with a low liquid injection coefficient of 1.4 g/Ah, both semisolid-state batteries and lithium-metal batteries passed the nail-penetration safety test, whereas the liquid electrolyte batteries failed, confirming the safety benefits of the solid-state electrolyte separator from the perspective of the cell configuration. The temperature-rise and voltage-drop curves of 8/9-layer pouch cells under different liquid injection coefficients indicate that as the liquid injection coefficient decreased, the temperature rise and voltage drop in the early stage before 200 °C were notably slower. For example, at a liquid injection coefficient of 1.8 g/Ah, the temperature rise rate was 163.4 °C/s, and the voltage drop rate was 3.52 V/s, whereas at a coefficient of 1.6 g/Ah, the corresponding values were 35.5 °C/s and 0.81 V/s. When the injection coefficient was further reduced to 1.4 g/Ah, the pouch cell passed the nail penetration test at only 0.02 °C/s and less than 0.01 V/s. Batteries with lower amounts of liquid electrolyte presented slower initial temperature increases and voltage decreases, indicating that

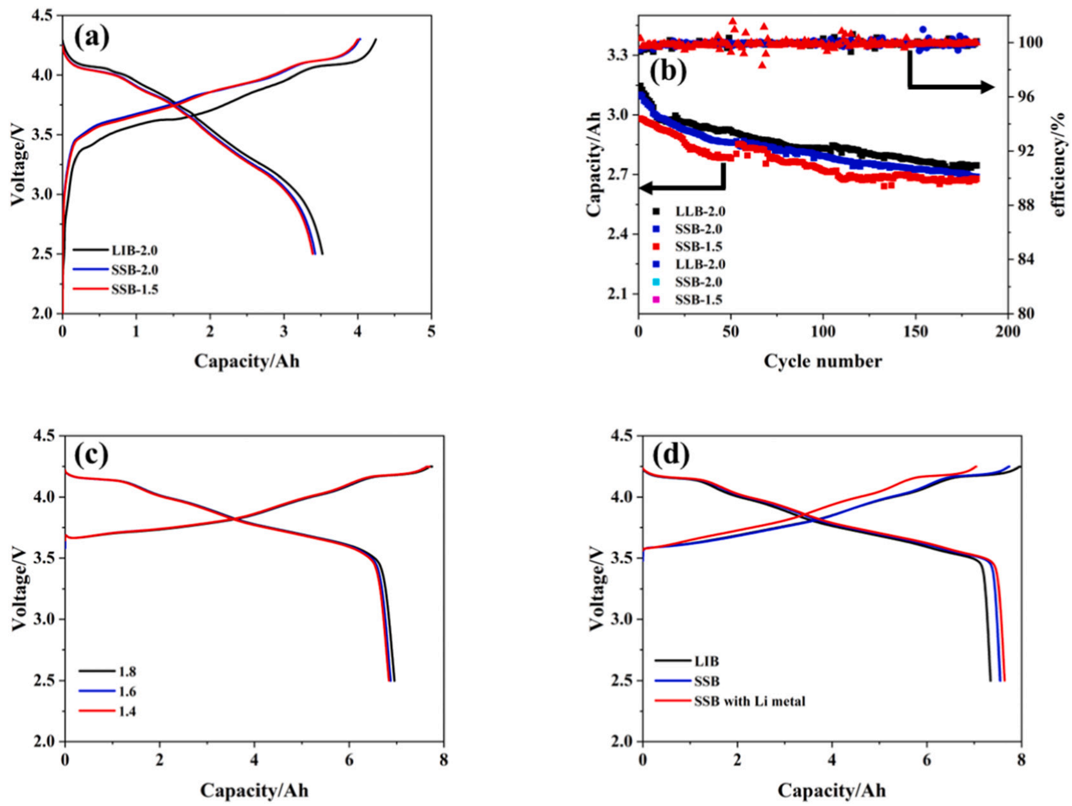

Fig. 3. Electrochemical performance comparison of pouch cells: (a, b) 3 Ah-level charge/discharge and cycling characteristics; (c) 7.5 Ah-level performance at different injection coefficients; (d) 7.5 Ah-level comparative analysis of battery types at 45 °C.

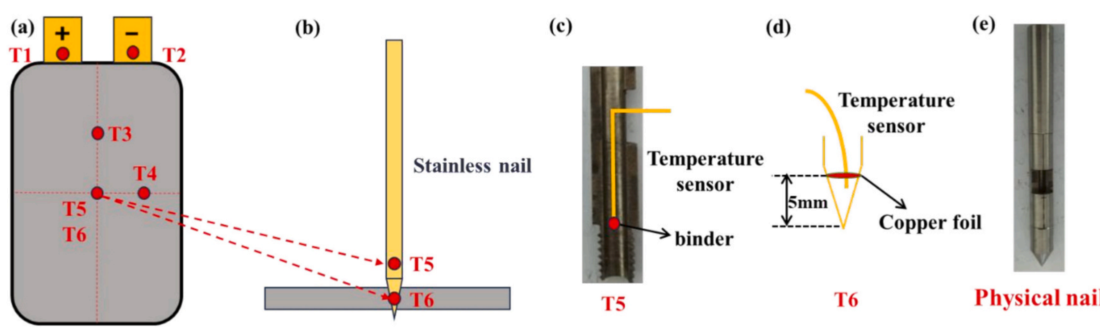

Fig. 4. Design of specialized stainless steel nail: (a, b) Temperature sensor (T1-T6) positioning during nail penetration; (c) T5 sensor array configuration; (d) T6 sensor placement diagram; (e) Physical prototype of the stainless steel nail.

reducing the liquid electrolyte amount can also delay the thermal spread of short circuits. Moreover, in 3/4-layer pouch cells, the lower liquid injection coefficients correspond to lower peak temperatures, further confirming this conclusion. These results verified the improvement effect of SSE separators on the safety of nail penetration through gradient testing from the perspectives of the cell configuration and amount of liquid electrolyte.

3.5. Mechanistic study of thermal runaway during nail penetration

To address the limitations of conventional monitoring tools such as the HIOKI 8431 instrument, which records only temperature and voltage fluctuations during nail penetration tests, we developed an innovative dual-battery parallel circuit model (Fig. 6(a)). This advanced system integrates Hall-effect coil current measurement technology (Fig. 6(b)) to enable comprehensive real-time monitoring of electrical parameters. By decomposing the complex nail penetration process, the model provides a quantitative analysis of critical thermal characteristics, including precise calculations of dynamic power variations and resistance changes throughout the penetration event. This methodological improvement significantly improves our understanding of battery failure mechanisms under mechanical abuse conditions.

To investigate the dynamic current characteristics during mechanical abuse, we established a parallel-connected pouch cell configuration using two fully charged cells with a matched voltage (4.2 V) and internal resistance (1.2 mΩ). The cells were connected via ultrasonic welding to minimize the ohmic contact resistance. Current monitoring was implemented via a Hall-effect coil (FS300E2) based on the physical transduction mechanism: short-circuit current → magnetic field → Hall voltage [50,51]. This galvanically isolated sensor was strategically positioned between the negative tabs to capture transient current variations with microsecond resolution. The experimental design uniquely enabled real-time current quantification during nail penetration: when one cell was mechanically punctured to induce internal short-circuiting,

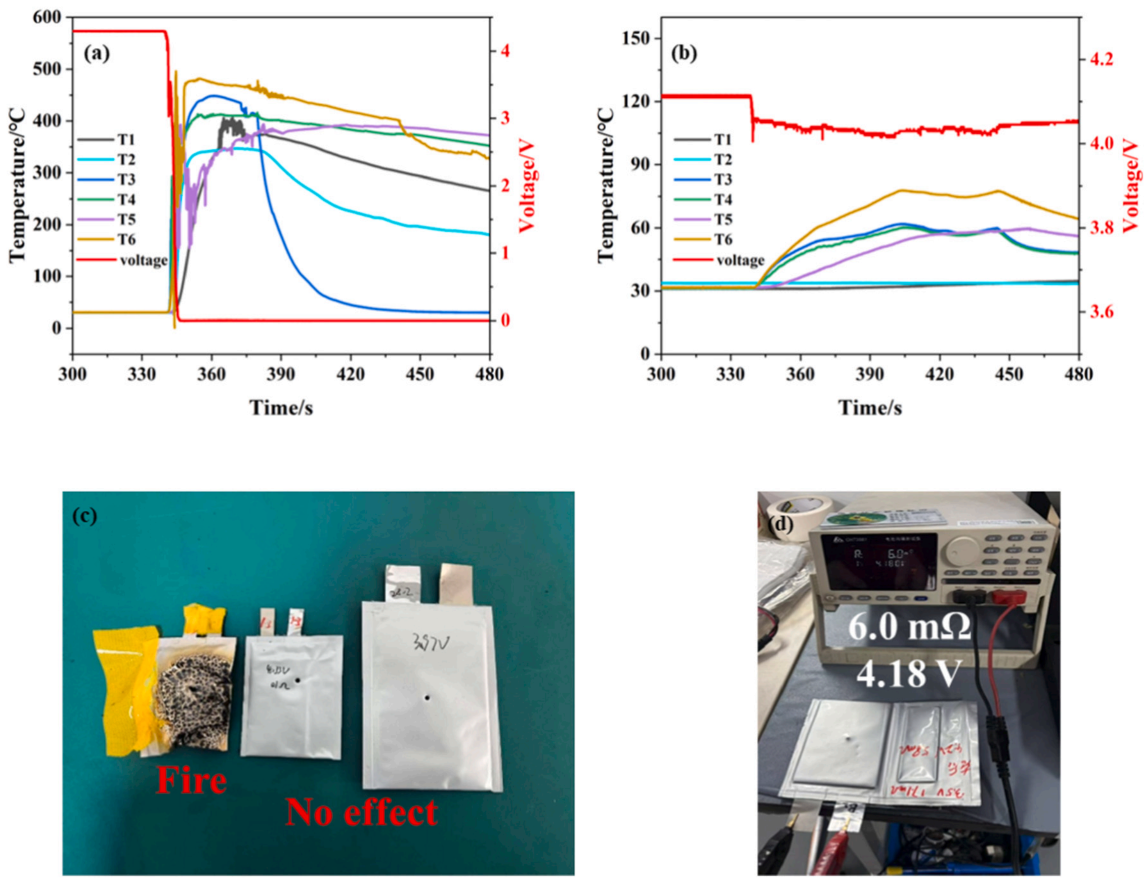

Fig. 5. Comparative analysis of thermal-electrochemical responses under nail penetration test: (a) Temperature/voltage evolution of liquid electrolyte battery during thermal runaway; (b) Stable thermal-electrochemical performance of semi-solid-state battery under identical test conditions; (c) Post-test visual comparison of battery structural integrity and debris characteristics; (d) Successful nail penetration validation of lithium metal semi-solid-state battery configuration.

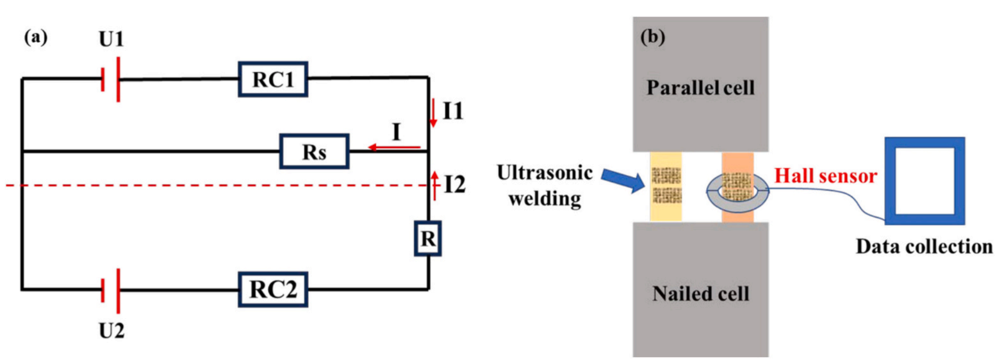

Fig. 6. Parallel circuit characterization: (a) Schematic representation of parallel-connected cell configuration; (b) Hall-effect current sensing methodology for parallel circuit analysis.

the parallel-connected cell was discharged to equalize the system voltage. The resultant current transients were monitored through the Hall sensor and visualized via custom-developed software with a 1 kHz sampling frequency. This innovative approach provides direct experimental validation of the equivalent circuit model's predictions regarding current redistribution during mechanical abuse events.

The developed parallel cell model demonstrates that the ultrasonic welding resistance between tabs is negligible compared with the internal resistance of the cell, as both cells (U1═U2, RC1 = RC2) exhibit identical electrochemical properties. Under this configuration, the current measured by the Hall-effect sensor (I2) directly corresponds to the nail penetration current (I1), establishing the relationship I1═I2. The model's series circuit configuration reveals that the total current through the penetration point (I) is the sum of individual currents (I=I1 + I2 = 2I2), enabling precise quantification of thermal power generation (P=U·I = 2 U2·I2) and short-circuit resistance (Rs = U/I=U/2I2). This approach effectively isolates mechanical abuse effects from electrochemical side reactions, providing a robust framework for analyzing internal short-circuit dynamics.

Validation experiments conducted on both liquid- and semisolid-state battery configurations revealed markedly different failure mechanisms during nail penetration tests. The liquid cell exhibited immediate thermal runaway, characterized by a rapid current surge to 49.8 A accompanied by catastrophic voltage collapse below 1 V (Fig. 7(b)). In

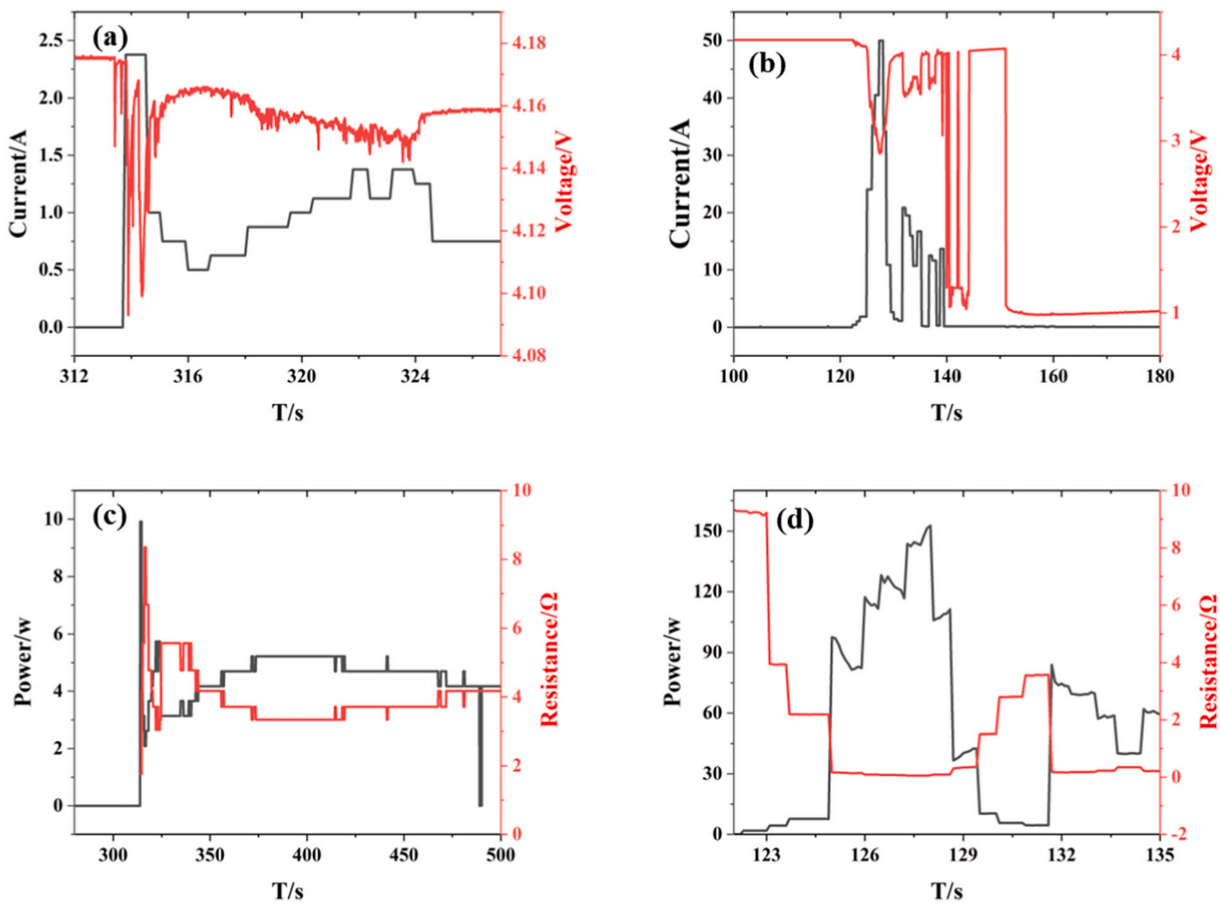

Fig. 7. Comparative analysis of electrical-thermal characteristics between stable and runaway battery systems: (a, c) Current-voltage profiles, heat dissipation power, and internal resistance evolution in non-thermal runaway batteries; (b, d) Corresponding parameters during thermal runaway progression.

stark contrast, the semisolid-state battery demonstrated remarkable stability, with the current gradually increasing to only 2.3 A and the voltage exhibiting minimal perturbation (0.08 V dip) before recovering to 4.16 V (Fig. 7(a)). These pronounced differences in electrical response patterns between battery types provide robust validation for the model's predictive capabilities. Further validation through multiple parallel-cell configurations confirmed the model's applicability across various experimental conditions. Using the established calculation framework, we systematically derived critical parameters, including the thermal power (P) and short-circuit resistance (Rs). Comprehensive data encompassing voltage, current, power, and resistance measurements are presented in Table 3, offering quantitative insights into the mechanical abuse behavior of different battery chemistries. The consistent correlation between the experimental observations and model predictions underscores its reliability for assessing battery safety under extreme conditions.

3.6. Disassembly characterization

The comparative dissection of nail-penetrated battery systems provided critical insights into the safety disparity between liquid and semisolid-state configurations. Short circuits originate from direct contact between the positive and negative electrodes, and the cause of thermal runaway is directly related to the contact area: the larger the short-circuit current is, the more heat is generated, and the higher the probability of thermal runaway [52]. The length of crack peaks along the nail penetration direction (height) and the resulting fracture area are closely related to nail penetration safety. Depth-of-field microscopy analysis of discharged liquid batteries revealed a minimal positive electrode fracture area of 2.3 mm^2 (Fig. 8 (a1)-(c1)), with longitudinal extension lengths consistently exceeding 4 mm (Fig. 8 (a2)-(c2)). In contrast, semisolid-state batteries exhibited distinct morphological changes: the positive electrode peak had an average length of only 2.23 mm (Fig. 8 (d2)-(f2)), and the fracture area caused by crack peaks was approximately 13 mm^2, coupled with the uniform encapsulation of the SSE separator at the penetration site (Fig. 8 (d1)-(f1)). This encapsulation mechanism effectively prevented direct positive–negative electrode contact, which was correlated with the observed elongation in the SSE separator (Table 1); then, sudden stress release caused the expansion of the fracture area and mitigated large-scale short-circuit formation. The combined effect of these factors leads to a smaller contact area in semisolid-state batteries, which significantly reduces the short-circuit current, lowers the risk of short-circuit-induced thermal runaway, and enhances nail penetration safety.

The safety advantages were further substantiated by two key factors: (1) the physical barrier properties of the SSE separator, which blocks large-scale short-circuit and exothermic reactions, and (2) the reduced liquid electrolyte amount in semisolid-state batteries, which minimizes heat accumulation and spread. These synergistic effects collectively suppressed thermal propagation pathways, providing a structural basis for the previously observed electrical stability during nail penetration. These findings conclusively demonstrate that semisolid-state architectures enhance mechanical abuse tolerance through both material properties and design optimization.

4. Conclusions

This work investigated the enhancement in nail penetration safety

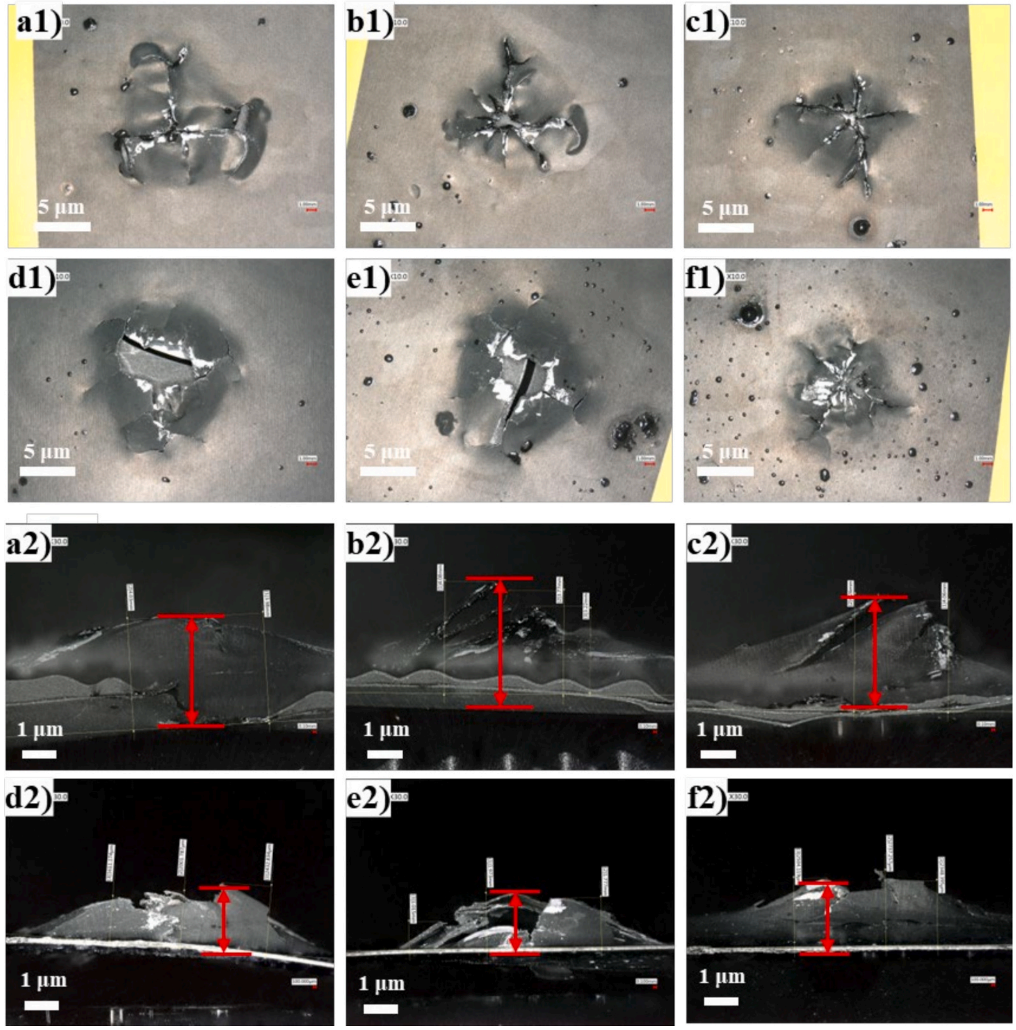

Fig. 8. High-resolution morphological comparison of electrode degradation patterns and puncture sites: (a1-c1) Sequential cross-sectional analysis of nail penetration path in liquid electrolyte battery; (d1-f1) Corresponding puncture track characterization in semi-solid-state battery configuration; (a2-c2) Spatially resolved positive electrode peak splitting in liquid system; (d2-f2) Distinctive electrochemical signature evolution in semi-solid-state counterpart.

Table 1
Comparative analysis of critical electrochemical parameters.

Sample type	Thickness (μm)	Areal density(g/m^2)	Tensile strength (Mpa)	Rate of elongation(%)	Permeability(Sec/100 mL)
Base separator	14	7.4	224	158	266.8
SSE separator	14	7.7	180	274	1698.5

provided by SSE separators compared with base separators. First, we developed and prepared SSE separators with different formulation ratios and selected the optimal one based on electrochemical performance for introduction and scale-up in pouch cells—the formulation with 50% PEO content proved to be the best. Second, nail penetration safety was studied under gradients of pouch cell capacity (stacking layer configuration) and liquid electrolyte injection amount (injection coefficient). The results validate the effectiveness of SSE separators in improving the nail penetration safety of batteries, which is closely related to high ductility and the ability to suppress electrode short-circuiting (by blocking electrode crack peaks). Reducing the liquid injection amount further inhibits the thermal spread upon short-circuiting, ultimately enabling a 450 Wh/kg semisolid-state lithium-metal battery to pass the nail penetration safety test at a liquid injection coefficient of 1.4 g/Ah. Finally, a quantitative analysis of the nail penetration process was conducted. A customized steel nail was used to accurately measure the internal temperature of the cell during penetration, and a parallel circuit model combined with a Hall-effect coil method was developed to quantify the short-circuit current during penetration. Parallel experiments verified the feasibility and reliability of this method. The short-circuit current of the semisolid-state battery using the SSE separator was only 1.5 A, whereas that of the liquid electrolyte battery using the base separator reached 87.45 A, indicating a significant difference. This reveals the beneficial effect of SSE separators in blocking short-circuiting during nail penetration and enhancing battery safety. These findings establish composite SSE separators as a viable solution for reconciling the safety–energy density trade-off in next-generation batteries.

CRediT authorship contribution statement

Xingyu Yang: Writing – original draft, Resources, Project

Table 2
Quantitative analysis of nail penetration test results.

Sample type	Capacity (Ah)	Cathode	Anode	Positive/ negative layers	Separator type	Electrolyte injection coefficient (g/Ah)	Voltage (V)	Weight (g)	Energy density (Wh/kg)	The max temperature of the nail/°C	Phenomena/ pass rate
Liquid battery	3.55	$LiNi_{0.93}Co_{0.5}Mn_{0.02}O_2$	SiOx/C 550	3/4	PP	2.0	3.64	36.7	352	>500	Fire/0
	3.27			3/4		1.5	3.64	34.8	342	>500	Fire/0
	7.34 (45 °C)			8/9		1.4	3.66	74.7	359	>500	Fire/0
Semi-solid-state battery	3.42			3/4	PEO + LLLZTO	2.0	3.65	35.7	351	31.2	No effect/1
	3.38			3/4		1.5	3.65	34.5	358	30.6	No effect/1
	6.97			8/9		1.8	3.66	75.9	336	>500	Fire/0
	6.95			8/9		1.6	3.66	74.8	340	>500	Fire/0
	6.84 (25 °C)			8/9		1.4	3.64	71.5	348	26.4	No effect/1
	7.54 (45 °C)						3.65		373		
	7.64 (45 °C)		Li	8/9		1.4	3.68	62.4	450	19.8	No effect/1

Table 3
Comparative peak parameter statistics across cell types.

Cell type	Max temperature of nail peak (°C)	Max voltage of nailed cell (V)	Max current of parallel cell (A)	Max discharge power (W)	Min resistance (Ω)
Liquid battery	>500	>4	49.8	418.32	0.042
	>500	>4	52.91	443.38	0.039
	>500	>4	87.45	731.082	0.024
Semi-solid-state battery	75	0.08	2.3	19.32	0.915
	66.2	0.06	1.52	12.54	1.395
	41.8	0.049	1.12	9.58	1.91

administration, Conceptualization. **Dong Liang:** Investigation, Formal analysis, Data curation, Conceptualization. **Tao Zhang:** Software, Resources, Investigation, Formal analysis. **Hailin Fan:** Validation, Resources, Investigation, Formal analysis. **Chongchong Zhao:** Software, Resources, Methodology, Investigation, Formal analysis. **Zhiwei Pan:** Visualization, Software, Investigation, Data curation. **Songwei Zhu:** Validation, Resources, Investigation, Formal analysis, Data curation. **Feng Huo:** Visualization, Supervision, Project administration, Funding acquisition. **Peigao Duan:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.est.2026.120847.

Data availability

Data will be made available on request.

References

[1] J.H. Kim, N.Y. Kim, Z.Y. Ju, Y.K. Hong, K.D. Kang, J.H. Pang, S.J. Lee, S.S. Chae, M. S. Park, J.Y. Kim, G.H. Yu, S.Y. Lee, Upscaling high-areal-capacity battery electrodes, Nat. Energy 10 (2025) 295–307, https://doi.org/10.1038/s41560-025-01720-0.
[2] Y.L. Liao, X.L. Wang, H. Yuan, Y.J. Li, C.M. Xu, S. Li, J.K. Hu, S.J. Yang, F. Deng, J. Liu, J.Q. Huang, Ultrafast Li-rich transport in composite solid-state electrolytes, Adv. Mater. 10 (2025) 2419782, https://doi.org/10.1002/adma.202419782.
[3] C.W. Park, J.H. Lee, J.K. Seo, W.Y. Jo, D. Whang, S.M. Hwang, Y.J. Kim, Graphene collage on Ni-rich layered oxide cathodes for advanced lithium-ion batteries, Nat. Commun. 12 (2021) 2145, https://doi.org/10.1038/s41467-021-22403-w.
[4] P.S. He, J.H. Park, Y.K. Jiao, R. Ganguli, Y.G. Huang, A. Lee, C.H. Ahn, M.N. Wang, Y.D. Peng, Y. Long, C.M. Chen, Z.H. Wang, Z.T. Tian, B.X. Mi, A.C. Arias, C. Fang, A. Toor, L.W. Lin, High-voltage water-scarce hydrogel electrolytes enable mechanically safe stretchable Li-ion batteries, Sci. Adv. 11 (2025) e3711, https://doi.org/10.1126/sciadv.adu3711.
[5] C.H. Lu, H.B. Jiang, X.R. Cheng, J.Q. He, Y. Long, Y.F. Chang, X.C. Gong, K. Zhang, J.X. Li, Z.F. Zhu, J.X. Wu, J.J. Wang, Y.Y. Zheng, X. Shi, L. Ye, M. Liao, X.M. Sun, B. J. Wang, P.N. Chen, Y.G. Wang, H.S. Peng, High-performance fiber battery with polymer gel electrolyte, Nature 629 (2024) 86–91, https://doi.org/10.1038/s41586-024-07343-x.
[6] J.H. Jang, C.Z. Wang, G.M. Kang, C.H. Han, J. Han, J.S. Shin, S. Ko, G. Kim, J. Baek, H.T. Kim, H. Lee, C.B. Park, D.H. Seo, Y.Z. Li, J.H. Kang, Miniature Li^+ solvation by symmetric molecular design for practical and safe Li-metal batteries, Nat. Energy 10 (2025) 502–512, https://doi.org/10.1038/s41560-025-01733-9.
[7] Q.K. Zhang, X.Q. Zhang, J. Wan, N. Yao, T.L. Song, J. Xie, L.P. Hou, M.Y. Zhou, X. Chen, B.Q. Li, R. Wen, H.J. Peng, Q. Zhang, J.Q. Huang, Homogeneous and mechanically stable solid–electrolyte interphase enabled by trioxane-modulated electrolytes for lithium metal batteries, Nat. Energy 8 (2023) 725–735, https://doi.org/10.1038/s41560-023-01275-y.
[8] X.P. Yi, Y. Yang, J.J. Song, L.Y. Gan, B.T. Wang, G.L. Jiang, K.S. Xiao, X.N. Song, N. Wu, L.Q. Chen, Hong Li, Strategically tailored polyethylene separator parameters enable cost-effective, facile, and scalable development of ultrastable liquid and all-solid-state lithium batteries, Energy Storage Mater. 77 (2025) 104191, https://doi.org/10.1016/j.ensm.2025.104191.
[9] T.Y. Hou, D.H. Wang, B.W. Jiang, Y. Liu, J. Kong, Y.B. He, Y.H. Huang, H.H. Xu, Ion bridging enables high-voltage polyether electrolytes for quasisolid-state batteries, Nat. Commun. 16 (2025) 962, https://doi.org/10.1038/s41467-025-56324-9.
[10] H.X. Liu, C. Liu, Y.H. Zhou, Y. Zhang, W.T. Deng, G.Q. Zou, H.S. Hou, X.B. Ji, The application of Al_2O_3 in separators and solid electrolytes of lithium-ion battery: A review, Energy Storage Mater. 71 (2024) 103575, https://doi.org/10.1016/j.ensm.2024.103575.
[11] K.S. Huang, C. Fang, J. Zhao, S.Q. Tian, X.G. Zhang, Solid electrolyte interfacial layer-modified ultra-thin separator facilitates the design of high specific energy lithium-metal batteries, ACS Appl. Eng. Mater. 1 (2023) 1775–1780, https://doi.org/10.1021/acsaenm.3c00171.
[12] H. Zhou, L. Huang, M. Yu, X.H. Ning, Stabilizing dual-cation liquid metal battery for large-scale energy storage: A comprehensive hybrid design approach, Energy Storage Mater. 70 (2024) 103540, https://doi.org/10.1016/j.ensm.2024.103540.
[13] J. Liu, Z.N. Bao, Y. Cui, E.J. Dufek, J.B. Goodenough, P. Khalifah, Q.Y. Li, B. Y. Liaw, P. Liu, A. Manthiram, Y.S. Meng, V.R. Subramanian, M.F. Toney, V. V. Viswanathan, M.S. Whittingham, J. Xiao, W. Xu, J.H. Yang, X.Q. Yang, J.

G. Zhang, Pathways for practical high-energy long-cycling lithium metal batteries, Nat. Energy 4 (2019) 180–186, https://doi.org/10.1038/s41560-019-0338-x.

[14] J. Chang, Q.Y. Huang, Y. Gao, Z.J. Zheng, Pathways of developing high-energy-density flexible lithium batteries, Adv. Mater. 33 (2021) 2004419, https://doi.org/10.1002/adma.202004419.

[15] J.H. Kim, K.M. Lee, J.W. Kim, et al., Regulating electrostatic phenomena by cationic polymer binder for scalable high-areal-capacity Li cell electrodes.J.H. Kim, K.M. Lee, J.W. Kim, S.H. Kweon, H.S. Moon, T. Yim, S.K. Kwak, S.Y. Lee, Regulating electrostatic phenomena by cationic polymer binder for scalable high areal-capacity Li battery electrodes, Nat. Commun. 14 (2023) 5721, https://doi.org/10.1038/s41467-023-41513-1.

[16] K.C. Pan, L. Zhang, W.W. Qian, X.K. Wu, K. Dong, H.T. Zhang, S.J. Zhang, A flexible ceramic/polymer hybrid solid electrolyte for solid-state lithium metal batteries, Adv. Mater. 17 (2020) 2000399, https://doi.org/10.1002/adma.202000399.

[17] X.H. Lu, M. Hao, Y.H. Fu, M.M. Xie, S.K. Su, Y.X. Hu, X. Bian, Y.H. Xie, Y.S. Ba, X. Y. Zhao, X.F. Du, Z.W. Chen, D.C. Ba, K. Liu, J, Metalized plastic current collectors: Bridge to high energy density reconciling high safety for lithium-ion batterie, J. Energy Storage 125 (2025) 116813, https://doi.org/10.1016/j.est.2025.116813.

[18] C. Liu, X.X. Zhang, J.X. Sheng, K. Wang, L.X. Yin, Y.Q. Shan, C.Q. Cao, P.G. Duan, D. Jia, K. Kapusta, Cohydrothermal carbonization of defatted microalgae and glucose for the production of supercapacitor carbon material, J. Energy Storage 80 (2024) 110258, https://doi.org/10.1016/j.est.2023.110258.

[19] S.Z. Damnali, K.M. Mazloumi, B. Tekin, T. Woehrle, A. Hoffmann, M. Hoelzle, Effect of carbon black properties on dry electrode processing of cathodes for lithium-ion batteries, J. Power Sources 646 (2025) 237243, https://doi.org/10.1016/j.jpowsour.2025.237243.

[20] L. Zhu, J.C. Chen, Y.W. Wang, W.L. Feng, Y.Z. Zhu, S.F.H. Lambregts, Y.M. Wu, C. Yang, E.R.H. Eck, L.M. Peng, A.P.M. Kentgens, W.P. Tang, Y.Y. Xia, Tunneling interpenetrative lithium ion conduction channels in polymer-in-ceramic composite solid electrolytes, J. Am. Chem. Soc. 10 (2024) 6591–6603, https://doi.org/10.1021/jacs.3c11988.

[21] Y. Wang, C.X. Zhu, R. Pfattner, H.P. Yan, L.H. Jin, S.C. Chen, F.M. Lopez, F. Lissel, J. Liu, N.I. Rabiah, Z. Chen, J.W. Chung, C. Linder, M.F. Toney, B. Murmann, Z. N. Bao, A highly stretchable, transparent, and conductive polymer, Sci. Adv. 3 (2017) 1602076, https://doi.org/10.1126/sciadv.1602076.

[22] S. Li, J.Z. Lu, Z. Geng, Y. Chen, X.Q. Yu, M. He, H. Li, Solid polymer electrolyte reinforced with a $Li_{1.3}Al_{0.3}Ti_{1.7}(PO_4)_3$-coated separator for all-solid-state lithium batteries, ACS Appl. Mater. Interfaces 14 (2022) 1195–1202, https://doi.org/10.1021/acsami.1c21804.

[23] K. Yu, H.P. Zeng, J. Ma, Y.D. Jiang, H.Y. Li, L.D. Zhang, Q.Q. Zhang, X.Y. Shan, T. T. Li, X.Q. Wu, H.L. Xu, W. Huang, C.Y. Wang, S.S. Chi, J. Wang, Q. Gong, Y. H. Deng, High-performance lithium metal batteries enabled by a nanosized garnet solid-state electrolyte modified separator, Chem. Eng. J. 480 (2024) 148038, https://doi.org/10.1016/j.cej.2023.148038.

[24] Y.M. Ren, H.R. Zhang, L. Wei, X.N. Feng, W.X. Wu, J, Multilayer internal short-circuit mechanism and thermal runaway risk assessment of lithium-ion batteries using nail penetration simulation, Energy Storage 97 (2024) 112824, https://doi.org/10.1016/j.est.2024.112824.

[25] Q.S. Wang, P. Ping, X.J. Zhao, G.Q. Chu, J.H. Sun, C.H. Chen, Thermal runaway caused fire and explosion of lithium-ion battery, J. Energy Chem. 93 (2024) 32–45, https://doi.org/10.1016/j.jpowsour.2012.02.038.

[26] X.N. Feng, D.S. Ren, X.M. He, M.G. Ouyang, Mitigating thermal runaway of lithium-ion batteries, Joule 4 (2020) 743–770, https://doi.org/10.1016/j.joule.2020.02.010.

[27] Y.S. Ye, L.Y. Chou, Y.Y. Liu, H.S. Wang, H.K. Lee, W.X. Huang, J.Y. Wan, K. Liu, G. M. Zhou, Y.F. Yang, A.K. Yang, X. Xiao, X. Gao, D.T. Boyle, H. Chen, W.B. Zhang, S. C. Kim, Y. Cui, Ultralight and fire-extinguishing current collectors for high-energy and high-safety lithium-ion batteries, Nat. Energy 5 (2020) 786–793, https://doi.org/10.1038/s41560-020-00702-8.

[28] J.G. Wang, W.X. Mei, Z.X. Cui, W.X. Shen, Q.L. Duan, Y. Jin, J.B. Nie, Y. Tian, Q. S. Wang, J.H. Sun, Experimental and numerical study on penetration-induced internal short-circuit of lithium-ion cell, App. Therm. Eng. 171 (2020) 115082, https://doi.org/10.1016/j.applthermaleng.2020.115082.

[29] Z.G. Zhang, Y.L. Qu, Y.D. Li, Y.Z. Song, X.Q. Wang, D.S. Ren, H. Huo, L. Wang, Y. Xia, X.M. He, Mechanical regulation of Ni-rich cathode to "the golden mean" toward safe Li-ion batteries during nail penetration, Energy Storage Mater. 73 (2024) 103802, https://doi.org/10.1016/j.ensm.2024.103802.

[30] Q. Zhao, S. Stalin, C.Z. Zhao, L. Archer, Designing solid-state electrolytes for safe, energy-dense batteries, Nat. Rev. Mater. 5 (2020) 229–252, https://doi.org/10.1038/s41578-019-0165-5.

[31] S. Lei, Z.Q. Zeng, Y.K. Wu, M.C. Liu, S.J. Cheng, J. Xie, Noncoordinating flame retardants with varied vapor pressures enabling biphasic fire-extinguishing electrolyte for high safety lithium-ion batteries, Chem. Eng. J. 463 (2023) 142181, https://doi.org/10.1016/j.cej.2023.142181.

[32] L. Li, J.L. Yang, R. Tan, W. Shu, C.T. John, Z.X. Zhang, Y. Zhao, C. Li, Y.J. Zhang, X. C. Li, H.Z. Zhang, X. Zhao, Z.K. Kou, Y. Xiao, F. Verpoort, H. Wang, L.Q. Mai, D. P. He, Large-scale current collectors for regulating heat transfer and enhancing battery safety, Nat. Chem. Eng. 1 (2024) 542–551, https://doi.org/10.1038/s44286-024-00103-8.

[33] A.S.J. Alujjage, B.S. Vishnugopi, A. Karmakar, D.P. Magee, Y. Barsukov, P. P. Mukherjee, Internal temperature evolution metrology and analytics in li-ion cells, Adv. Funct. Mater. (2025) 2417273, https://doi.org/10.1002/adfm.202417273.

[34] J.B. Fan, C.C. Liu, N. Li, L. Yang, X.G. Yang, B.W. Dou, S.J. Hou, X.N. Feng, H. Q. Jiang, H. Li, W.L. Song, L. Sun, H.S. Chen, H.J. Gao, D.N. Fang, Wireless transmission of internal hazard signals in Li-ion batteries, Nature 641 (2025) 639–645, https://doi.org/10.1038/s41586-025-08785-7.

[35] P.F. Lv, D. Zhang, H. Shi, Y. Lin, S. Xie, M.Y. Zhi, Y.H. He, C.Y. Tang, Polyethylene ceramic separators with semi-interpenetrating polymer network boosting fast-charging cycle capacity retention and safety for lithium-ion batteries, ACS Appl. Energy Mater. 7 (2024) 13–20, https://doi.org/10.1016/j.jpowsour.2023.233022.

[36] B.B. Mao, H.D. Chen, Z.X. Cui, T.Q. Wu, Q.S. Wang, Failure mechanism of the lithium-ion battery during nail penetration, Int. J. Heat Mass Transf. 122 (2018) 1103–1115, https://doi.org/10.1016/j.ijheatmasstransfer.2018.02.036.

[37] Q.S. Wang, P. Ping, X.J. Zhao, Thermal runaway caused fire and explosion of lithium-ion battery, J. Power Sources 208 (2012) 210–224, https://doi.org/10.1016/j.jpowsour.2012.02.038.

[38] H. Uwitonze, A. Ni, V.M. Nagulapati, H. Kim, H. Lim, CFD study of nail penetration induced thermal runaway propagation in lithium-ion battery cell pack, Appl. Therm. Eng. 243 (2024) 122649, https://doi.org/10.1016/j.applthermaleng.2024.122649.

[39] G.Z. Liang, Y.M. Zhang, Q. Han, Z.P. Liu, Z. Jiang, S. Tian, A novel 3D-layered electrochemical-thermal coupled model strategy for the nail-penetration process simulation, J. Power Sources 342 (2017) 836–845, https://doi.org/10.1016/j.jpowsour.2017.01.005.

[40] B. Liu, S. Yin, J. Xu, Integrated computation model of lithium-ion battery subject to nail penetration, Appl. Energy 183 (2016) 278–289, https://doi.org/10.1016/j.apenergy.2016.08.101.

[41] D. Chaykina, M. Ghosh, O.s Kudu, Critical outlook on separator layers for solid-state lithium batteries: Solid electrolyte materials, anode interface engineering, & scalable separator production, J. Power Sources 643 (2025) 237014, https://doi.org/10.1016/j.jpowsour.2025.237014.

[42] G. Zhou, C.X. Niu, X. Jiang, M.Q. Zhang, J.J. Zhao, Q. Zhang, Lithium-ion battery safety polymer electrolyte membrane based on PVDF-HFP prepared by electrospinning method, J. Energy Storage 123 (2025) 116734, https://doi.org/10.1016/j.est.2025.116734.

[43] D.L. Chen, Z.H. Wang, M. Zhu, X.P. Yang, G. Sui, In situ covalently bridged ceramic-polymer electrolyte with fast, durable ions conductive channels for high-safety lithium batteries, J. Colloid Interface Sci. 686 (2025) 126–135, https://doi.org/10.1016/j.jcis.2025.01.198.

[44] A. García, J.M. Serrano, J.M. Gimeno, C.G. Figueira, Comparative CT scan and U-net segmentation analysis of NMC811 batteries after thermal runaway triggered by various abuse methods and states of charge, J. Power Sources 629 (2025) 236045, https://doi.org/10.1016/j.jpowsour.2024.236045.

[45] H. Kim, H.J. Seifert, C. Ziebert, P. Finster, J. Friedl, A comparative study of commercial automotive prismatic Li-ion cells using nail penetration test, differential scanning calorimetry and thermogravimetric analysis, J. Power Sources 623 (2024) 235416, https://doi.org/10.1016/j.jpowsour.2024.235416.

[46] H. Wu, D. Zhuo, D.S. Kong, Y. Cui, Improving battery safety by early detection of internal shorting with a bifunctional separator, Nat. Commun. 5 (2014) 5193, https://doi.org/10.1038/ncomms6193.

[47] L.S. Lin, J.L. Li, I.M. Fishman, L. Torres-Castro, Y. Preger, V. De Angelis, J. Lamb, X. Q. Zhu, S. Allu, H. Wang, Mechanically induced thermal runaway severity analysis for Li-ion batteries, J. Energy Storage 61 (2023) 106798, https://doi.org/10.1016/j.est.2023.106798.

[48] M. Li, J. Lu, Z. Chen, K. Amine, 30 years of lithium-ion batteries, Adv. Mater. 30 (2018) 1800561, https://doi.org/10.1002/adma.201800561.

[49] X. Chen, Q.S. Sun, J. Xie, C. Huang, X.W. Xu, J. Tu, X.B. Zhao, T.J. Zhu, Double-layer solid-state electrolyte enables compatible interfaces for high-performance lithium metal batteries, J. Energy Chem. 74 (2022) 91–99, https://doi.org/10.1016/j.jechem.2022.07.006.

[50] Y. Sharon, B. Khachatryan, D. Cheskis, Sensor, toward a low current hall effect sensor, Actuat. A-Phys. A 279 (2018) 278–283, https://doi.org/10.1016/j.sna.2018.06.027.

[51] J.L. Liu, J.L. Liu, X.M. Jin, Y. Zhang, J.H. Gong, Z.R. Wang, Y.Y. Cui, Influence of multi factors on external short circuit of lithium-ion battery, Process Saf. Environ. 199 (2025) 107279, https://doi.org/10.1016/j.psep.2025.107279.

[52] D.S. Ren, X.N. Feng, L.S. Liu, H. Hsu, L.G. Lu, L. Wang, X.M. He, M.G. Ouyang, Investigating the relationship between internal short circuit and thermal runaway of lithium-ion batteries under thermal abuse condition, Energy Storage Mater. 34 (2021) 563–573, https://doi.org/10.1016/j.ensm.2020.10.020.